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Ensemble Learning based Classification of Edible and Poisonous Agaricus Mushrooms

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***Abstract*— Agriculture has a crucial role in sustaining human populations in numerous countries globally, particularly in economies that are still in the process of development. Mushroom growing has emerged as a prominent agricultural practice in contemporary times. The primary and essential element in the practice of mushroom growing is the identification of mushroom damage. Mushrooms have the potential to be contaminated by various microorganisms such as bacteria, pests, and pathogenic molds. These contaminants can lead to significant crop losses and pose health risks to consumers. Mushrooms can be classified into two categories based on their health implications: those that are deemed safe for consumption, referred to as healthy or edible mushrooms, and those that pose a risk to human health if ingested, known as unhealthy or dangerous mushrooms. Various strategies for mushroom categorization exist; however, these methods are predominantly manual and can be time-consuming, particularly when dealing with large quantities of items that need to be categorised. The utilization of machine learning and deep learning technologies has grown prevalent in the classification of agricultural products. This study involved an analysis of 8124 data samples from the UCI repository Mushroom dataset for the purpose of classification. This dataset was originally contributed by The Audubon Society Field Guide to North American Mushrooms. This study employs various approaches, namely Random Forest, Bagging, Gradient Boosting, Adaboost, and XG Boost. Among the several approaches considered, Adaboost exhibited the highest performance, achieving an accuracy of 95.35% in a 10-fold cross-validation.**

Keywords—Mushroom, poisonous, Bagging, Classification, Boosting, ensemble learning.

I. INTRODUCTION

Mushrooms are the fleshy and soft fruiting structures, belong to a fungus family and are rich in vitamins and minerals. Agaricus mushroom is a popular species of mushroom which is cultivated in various countries (approx. 70)[1] and is the most frequently and widely used mushroom in India also. These mushrooms have gradually grown in popularity due to their delicious flavour, wide supply, and desirability as a healthy food. Mushrooms are low in fat, cholesterol, sodium and calories, but high in protein, vitamin B, vitamin D, and, vitamin K, carbohydrate, important amino acids, and fibers[1],[2]. As a result, consumption has expanded, and mushroom cultivation has become more industrialised. But while cultivating mushrooms, because of bacteria, pests, unfavourable temperature, soil conditions and pathogenic molds, mushrooms can be healthy(edible) and unhealthy (poisonous). All of this can cause major crop failure[3]. These types of mushrooms may contain viruses that carry infectious diseases. Agaricus Mushrooms are prone to a number of bacterial infections, including drippy gills, brown blotch, soft rot, and internal stipe necrosis diseases, all of which have a negative impact on mushroom quality[4]. Pseudomonas tolaasii Paine bacteria has long been thought to be the primary origin of the mushroom disease[4].

In appropriate circumstances or when triggered by exudates from top of the mushroom, these bacteria release a final product of their metabolism called tolaasin[4]. This chemical can be poisonous to Agaricus mycelium in high enough concentrations. Mushrooms must be characterized as healthy or unhealthy in order to ensure proper mushroom crop management and maximum crop production. An assessment framework must be created to verify that only good quality mushrooms reach the customers and that unhealthy regions of the field are effectively isolated.

Toxic mushroom consumption can make humans sick and may even cause death. Abdominal pain, nausea, vomiting, stomach discomfort, and diarrhoea are the most common gastrointestinal symptoms caused by toxic mushrooms[5]. As per the survey conducted in the year 2013 to 2017, by the Disease Control Department, Bureau of Epidemiology, Every year 1,200 people got health issues after eating mushrooms and 4 died by consuming poisonous mushrooms. In 2015, a

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report from the Bureau of Epidemiology, stated that 100,000 people, or 2.10 percent of the population, suffered from hazardous effects after eating poisonous mushrooms[5]. The major factor of hazardous mushroom intake is insufficient information and expertise in distinguishing between healthy and unhealthy mushrooms.

A few poisonous mushrooms are a vital piece of natural medication. Medical experts are using amatoxins properties from amanita mushrooms to regulate and treat cancers nowadays. Amanita muscaria is a well-known mushroom species that has been used to generate sedative and anaesthetic medications [6]. Toxins present in Amanita muscaria and albicans species are helpful in capturing and killing the agricultural insects and pests which are responsible for crop diseases. In this way they are very important for biological management. Hence, effective identification and classification of unhealthy mushrooms is required, both in terms of toxicity avoidance and use of these poisons.

Some methods which are used currently for identifying poisonous mushrooms are divided into four types: fungal classification, chemical determination, folk experience and animal experimentation[7]. Currently, research into toxic mushrooms using these approaches is not only inconclusive, but also leaves a lot to be desired. In today 's rapid scientific innovations, Computer `intelligence has enabled us to detect diseases more precisely than it has ever been. One of them is data mining, which may be utilized to make accurate predictions. It's a method for extracting additional valuable information from previously collected data by examining it. Forecasting is heavily reliant on machine learning.

In machine learning, there are numerous classification strategies available. For mushroom classification, some research has been done previously. Existing research are based on Logistic Regression, ANN, Decision Tree (C4.5), HSI, ZeroR, SVM (Support Vector Machine), Bayes Net, Gaussian Naïve Bayes, and KNN classifier algorithms[8],[9],[10],[11]. However, ensemble methods give better outcomes than existing single classifier techniques. In ensemble learning technique, the predictions of weak models are combined, hence, it performs better than single classifiers[12].

In this study, we used three approaches of ensemble learning to classify the mushrooms: random forest, bagging, and boosting. Bagging is done with decision tree and KNN algorithm, random forest is done with decision tree, and boosting is done with Adaboost, Gradient boost and XGBoost. In all base classifiers, the feature sets are randomly picked for classification tasks. The goal of this study is to develop the optimal ensemble approach for classifying healthy and unhealthy mushrooms with the maximum efficiency and minimum failure rate.

II. RELATED WORKS

Identifying and classifying poisonous mushrooms has long been a challenging and time-consuming process. Many researchers have looked into traditional approaches as well as computational technologies for identifying edibility of mushrooms. This section of the paper goes on some of the approaches that were used.

Wang and Yingying proposed a model for toxic mushroom identification by using the Multi Grained Cascade Forest Classifier , Logistic Regression and SVM method. The mushroom samples they have used is from the University of California, and for missing value they applied KNN algorithm. gcForest is an ensemble approach which uses decision tree. It has two stages: one is Multigrained screening, which creates the features of data sample, and second is the cascade forests, which provide prediction probability outcomes using numerous forest cascades. Using the Multi Grained Cascade Forest Classifier, they were able to attain 98 percent accuracy, with an accuracy variance of 8% [7].

For recognition of poisonous Amanita mushrooms, Wang and Peng designed a new convolutional neural network called B-CNN, where B stands for Bilinear. An attention mechanism model is applied with B-CNN. They used their own created dataset which contains 3219 images of Amanita mushroom. To reduce the model size and minimize the training time they added a Convolutional attention mechanism (CBNM) with this model and compared it with VGG, ResNet, EfficientNet. The model has accuracy of 95.2% in classification of 7 different species of Amanita[6].

Tongcham and Phongsakhon developed a method to detect contaminated oyster mushroom spawns. Test samples of mushroom spawn were taken from a field in Thailand. Firstly Spawn sample ROI (regions of interest) were retrieved and processed to decrease distortion. After that Trivariate histograms were created from these locations and taken as a classification feature. They performed a comparison between 5 methods deep neural

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network (DNN), k-nearest neighbor (KNN), support vector machines (SVMs), decision trees, and nearest centroid classifier (NCC); DNN got the highest accuracy 98.8% among them[13].

Chaudhary and Archana presented hybrid ensemble methods for multiclass classification problems in a dataset of oilseed disease. The classification test contains 4 common crop data samples: mushroom, Soybean, Grub-damage and Iris plants. They combined Naïve Bayes and Logistic Regression with the help of ensemble-Vote. To achieve better results they applied a combination rule with ensemble vote. After applying 10-fold cross validation they got 94.73% accuracy [14].

Ismail studied the physical behaviour of mushrooms, which contains size, roots, surface, Gill shape and size, color of the cap, stems and many more. They applied the principal component analysis (PCA) to choose the optimal features for the categorization. They used K-Nearest Neighbour, Naive Bayes and decision tree (DT) algorithm (J48 classifier) to build a prediction model for the mushroom dataset from the UCI Machine Learning Repository[15].

III. METHODOLOGY

We have selected five well-known Ensemble techniques. Ensemble learning is a powerful predictive model which contains multiple algorithms (known as "weak learners" or "base learners") that are firstly trained and then combined together to provide the correct prediction for a given issue[16]. We can get a superior result if we integrate multiple base models appropriately and precisely [17]. For classification and prediction, the Ensemble ML method is effective in providing correct results.

A. Data Collection

The mushroom dataset was collected from UCI machine learning repository. Dataset is uploaded by Jeff Schlimmer and originally it was obtained from *The Audubon Society Field Guide to North American Mushrooms*. There are 8124 samples in the mushroom dataset, 22 features or attributes, and two class categories that are poisonous and edible. In 8124 data samples, 4208 are edible and 3916 are poisonous. The attributes are bruises, cap shape, odor, cap surface, gill spacing, cap color, gill attachment, veil color, stalk color above ring, gill size, stalk shape, gill color, stalk surface above ring, stalk root, stalk surface below ring, population, veil type, stalk color below ring, habitat, spore print color ring type, and ring number.

TABLE I. DESCRIPTION OF AGARICUS MUSHROOM DATASET

	Agaricus mushroom dataset record	
	Total Features	***Total Records***
Data of Agaricus mushroom	22	8124

TABLE II. ATTRIBUTES AND THEIR MEANINGS

Attribute	**Attribute meaning**
Bruises	Yes , No
Cap shape	Conical, Sunken, Convex, Flat, Knobbed, Bell
Odor	Anise, Anise, Pungent, Foul , Spicy ,Musty, None,
Cap surface	Scaly, Fibrous , Smooth, Grooves
Gill spacing	Close , Distant, Crowded
Cap color	Buff, Pink, Cinnamon, Green, Brown, Purple, Gray, Red, Yellow, White
Gill attachment	Free, Attached, Notched, Descending
Veil color	White, Brown, Yellow, Orange
Stalk color above ring	Gray, Brown, Pink, Buff, White, Cinnamon, Yellow, Orange, Red
Gill size	Narrow, Broad
Stalk shape	Tapering, Enlarging
Gill color	Yellow, Green, Black, Pink, Brown, Red, Orange, Buff, Gray Chocolate, Purple, White
Stalk surface above ring	Silky, Fibrous, Smooth, Scaly,
Stalk root	Rooted, Bulbous, Equal, Club, Rhizomorphs, Cup, Missing
Stalk surface below ring	Silky, Fibrous, Smooth, Scaly
Population	Scattered, Abundant, Several, Clustered, Solitary, Numerous
Veil type	Universal, Partial
Stalk color below ring	Brown, Buff, Cinnamon, Gray, Orange, Pink, Red, White, Yellow
Habitat	Meadows, Grasses, Urban , Leaves, Waste, Paths, Woods
Spore print color	Orange, Black, White, Brown, Purple, Buff, Green, Chocolate, Yellow
Ring number	One , Two, None
Class	Toxic, Non-toxic

B. Data Preprocessing

We analyzed the Agaricus Mushroom data and transformed them into the appropriate format in order to extract more usable information, which has a direct impact on machine learning models capacity to learn and improve efficiency.

1.Importance- Graph analysis

An importance graph is a visual representation of the significance of attributes that may be used to accurately perform the prediction of the final class. By using the feature importance method we can assign a score to input features, depending on how relevant they are in predicting outcome values.

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Feature importance values contribute significantly in predictive analysis programs because they provide overview of the data, model understanding, and a framework for feature extraction which will improve the predictability and effectiveness of a classification method. Figure 1-5 depicts the importance graph of the implemented algorithms.

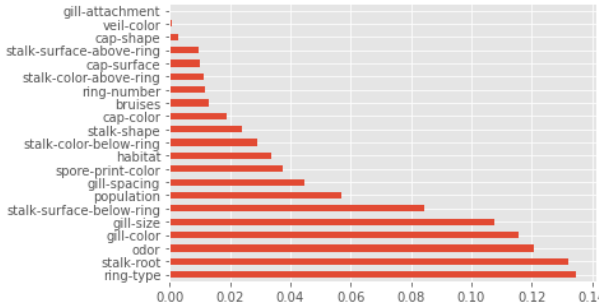

Fig. I. Importance- Graph of Random Forest

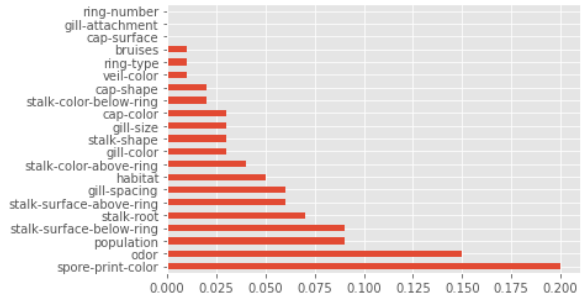

Fig.II. Importance- Graph of AdaBoost

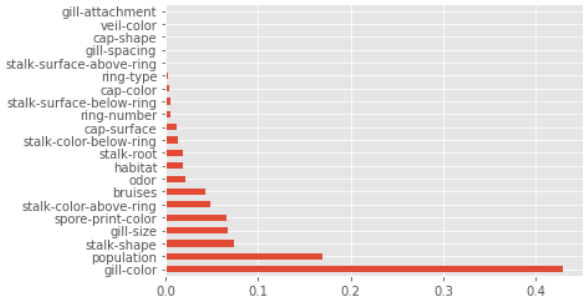

Fig. III. Importance- Graph of XGBoost

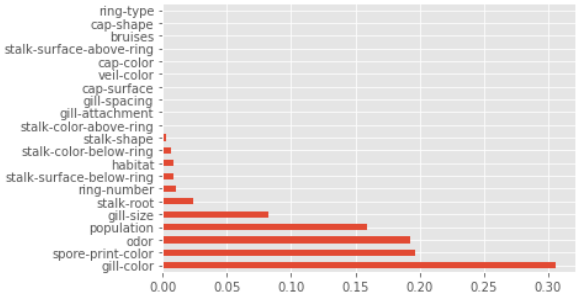

Fig. IV. Importance- Graph of Gradient Boost

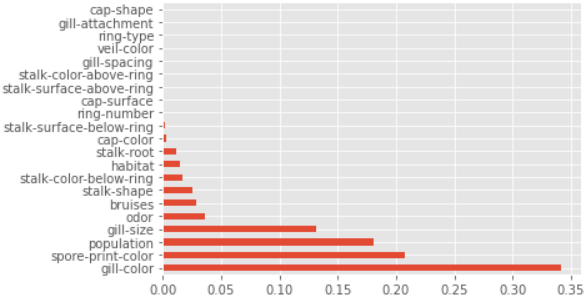

Fig. V. Importance- Graph of Bagging

2. Feature Selection using Filter Method

A variety of strategies can be employed to choose features, one of which is the filter approach. In filter technique, a correlation matrix is required for screening and usually, this matrix is obtained by using Pearson correlation[18]. We generated a Pearson correlation heatmap to visualize the relationship between attributes and the output class. Following the analysis of feature importance, two of the less important features, veil color, and gill attachment are removed from the original data.

C. Training Stage

Before applying the algorithm the dataset has been divided into training dataset and testing dataset. We splitted our dataset as 7:3 ratio for training and testing.

D. *Proposed System Architecture*

The framework of proposed system design is depicted in Figure 6. We used the UCI mushroom dataset, which contains information about physical features of the mushroom. To obtain an efficient approach, the Random Forest, XGBoost, Adaboost, Gradient Boosting and Bagging algorithms are implemented on the mushroom dataset. The purpose of this research is to find the most efficient mushroom categorization system.

E. Applied Algorithm

In this paper, five classification algorithms are applied on the UCI Agaricus Mushroom dataset for the healthy and unhealthy mushroom classification.

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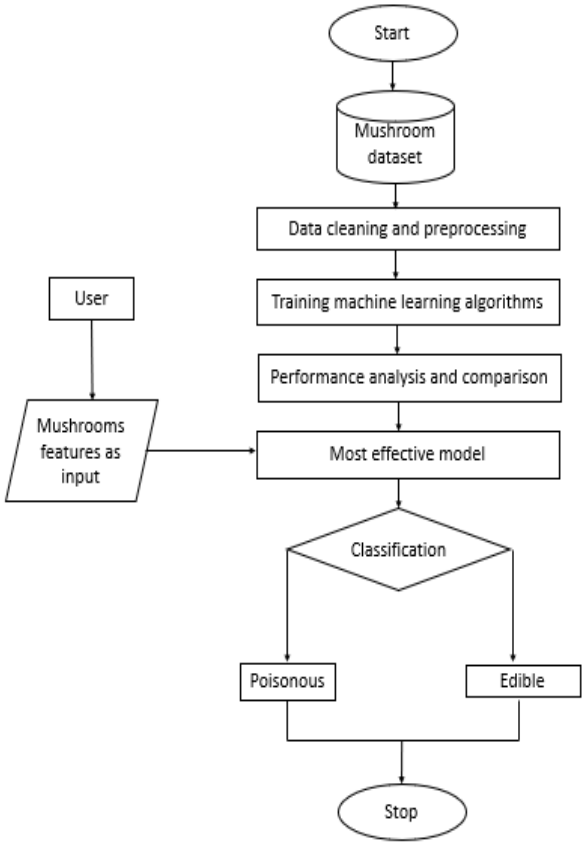

Fig. VI. Proposed system architecture

a) Random Forest Algorithm

RF is a decision tree-based algorithm. Firstly in the year 2001, Breiman introduced Random Forest, which merges Bagging algorithm with random feature selection. In RF, the result is directly proportional to the quantity of trees present in the forest. If more trees are present, we will receive more precise results. RF employs two classifiers J48 and C4.5[18].

b) *Bagging Algorithm*

Bagging is the primary ensemble learning strategy. It is broadly used among all the available ensemble learning algorithms. For every model in Bagging, a distinct arbitrarily selected sample of the original dataset is used. Every sample predicts a class, and the final output class determined by majority vote. The suggested bagging approach uses Decision tree and Knn algorithm as a weak learner method[19].

c) *Gradient Boosting Algorithm*

Gradient boosting(GB) is a popular boosting strategy. In gradient boosting, each model rectifies the inaccuracy of the previous model. The training sample weights are not modified, as they are in Adaboost; instead, every model is trained using the predecessor's faults[20].

d) *XG Boost Algorithm*

Boosting is a technique that introduces new strong models to adjust the errors of existing models. XGBoost is a decision tree ensemble approach, based on gradient boosting. It is a highly scalable method for classification. XGBoost optimizes the loss function and produces an additional extension of the objective function same as gradient boosting. The focus of XGBoost is on minimising the computational cost of choosing the appropriate split, it is the major time taking step of decision tree construction[20].

e) AdaBoost

AdaBoost is an acronym of "Adaptive Boosting". It is a machine learning methodology Where preceding classifier's inaccuracy affects the succeeding weak learners. This algorithm acts as an outlier and is robust to overfitting, as well as it is very sensitive to distorted data. [18],[21].

IV. RESULT ANALYSIS

We implemented ensemble learning methods on both processed and unprocessed mushroom data. For classification of healthy and unhealthy mushrooms on processed data, we got the test accuracy of 96.01 percent with Adaboost, 92.98 percent with GB, 87.97 percent with Bagging, 86.99 percent with RF, and 90.97 percent with XG Boost. Using the AdaBoost, we were able to attain the best outcomes.

TABLE III. TEST SCORE

	Ensemble learning methods				
	Random Forest	Bagging	Adaboost	Gradient Boosting	XG Boost
With selected features	86.99%	87.97%	96.01%	92.98%	90.97%
With all features	75.33%	76.89%	82.92%	81.53%	79.89%

A. Cross Validation Analysis

To examine how our classifiers responded to new data samples that were not used in the training phase, we employed the 10-fold cross validation process. Table IV represents the average of the cross validation result for all five methods. In 10-fold cross validation, while using the Adaboost,

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we were able to reach the greatest result of 95.35 percent.

TABLE IV. CROSS VALIDATION RESULT

	Ensemble Learning Techniques				
	RF	***Bagging***	***Adaboost***	***GB***	***XG Boost***
Average 10 fold Cross validation value	84.08%	85.79%	95.35%	90.06%	88.51%

B. *Analysis using Confusion Matrix*

An M x M matrix, called confusion matrix, is created to calculate the accuracy of an applied model. Here M denotes the target group. This matrix shows a comparison between predicted value and actual value for each model. With the help of a confusion matrix, we are able to analyse how our trained model is functioning and what errors it may generate. There are four terminologies to understand confusion matrix:

True Positive (TP): Model predicted Mushroom is poisonous, and actually it is.

True Negative(TN): Model predicted Mushroom is not poisonous (it's edible), and actually it's not.

False Positive (FP): Model predicted Mushroom is poisonous, but it is not.

False Negative (FN): Model predicted Mushroom is not poisonous, but it is actually poisonous.

TABLE V. CONFUSION MATRIX ANALYSIS

Models	Accuracy	Precision	Recall	F1 score
Random forest	86.99%	89.18%	86.25%	87.55%
Bagging	87.97%	85.29%	91.45%	88.26%
Adaboost	96.01%	95.82%	96.64%	96.22%
Gradient Boost	92.98%	95.14%	98.14%	96.61%
XG Boost	90.97%	91.27%	93.22%	92.23%

V. CONCLUSION

Ongoing study is being conducted on the identification of poisonous mushrooms, as well as the occurrence of associated fatalities and illnesses. Hence, the development of a computational model for the identification of toxic mushrooms is vital.. In this paper, five different ensemble learning methods are applied to classify Healthy and unhealthy Agaricus Mushrooms on the basis of its physical properties. Ensemble models can minimize the errors generated by single models. As a result performance is superior to that of the single best classifier. Here Adaboost shows the highest accuracy 95.35% with 10 fold cross validation. In the future, ensemble strategy can be expanded to cover different sorts of diseases that may be found on various types of plants, crops, and vegetables in real-time experimental research.

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